

Лабораторная Работа №5

Эмуляция и измерение потерь пакетов в глобальных сетях

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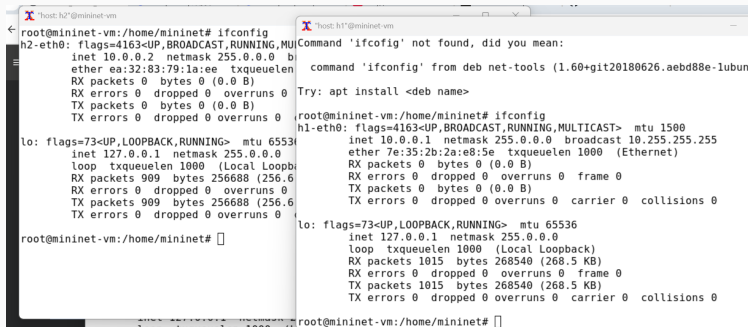
Выполнение лабораторной работы

Основной целью работы является получение навыков проведения интерактивных экспериментов в среде Mininet по исследованию параметров сети, связанных с потерей, дублированием, изменением порядка и повреждением пакетов при передаче данных. Эти параметры влияют на производительность протоколов и сетей.

Задание

1. Задайте простейшую топологию, состоящую из двух хостов и коммутатора с назначенной по умолчанию mininet сетью 10.0.0.0/8.
2. Проведите интерактивные эксперименты по исследованию параметров сети, связанных с потерей, дублированием, изменением порядка и повреждением пакетов при передаче данных.
3. Реализуйте воспроизводимый эксперимент по добавлению правила отбрасывания пакетов в эмулируемой глобальной сети. На экран выведите сводную информацию о потерянных пакетах.
4. Самостоятельно реализуйте воспроизводимые эксперименты по исследованию параметров сети, связанных с потерей, изменением порядка и повреждением пакетов при передаче данных. На экран выведите сводную информацию о потерянных пакетах.

Задал простейшую топологию, просмотрел ifconfig



The image shows two terminal windows side-by-side. The left window is titled 'host: h2@mininet-vm' and shows the output of the 'ifconfig' command for host h2. It displays details for the 'eth0' interface (IP 10.0.0.2, MTU 1500) and the 'lo' loopback interface (IP 127.0.0.1, MTU 65536). The right window is titled 'host: h1@mininet-vm' and shows an error message 'Command 'ifconfig' not found, did you mean:' followed by a suggestion to install the package. Below this, it shows the output of 'ifconfig' for host h1, displaying details for the 'eth0' interface (IP 10.0.0.1, MTU 1500) and the 'lo' loopback interface (IP 127.0.0.1, MTU 65536).

```
root@mininet-vm:/home/mininet# ifconfig
h2-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.2 netmask 255.0.0.0 broadcast 10.255.255.255
    ether ea:32:83:79:1a:ee txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 909 bytes 256688 (256.6 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 909 bytes 256688 (256.6 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet#
```

```
Command 'ifconfig' not found, did you mean:
  command 'ifconfig' from deb net-tools (1.60+git20180626.aebd88e-1ubuntu1)

Try: apt install <deb name>

root@mininet-vm:/home/mininet# ifconfig
h1-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.1 netmask 255.0.0.0 broadcast 10.255.255.255
    ether 7e:35:2b:2a:e8:5e txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1015 bytes 268540 (268.5 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1015 bytes 268540 (268.5 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet#
```

Figure 1: ifconfig

```
root@mininet-vm:/home/mininet# ping -c 6 10.0.0.1
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data.
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=1.42 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=0.056 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=0.045 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=0.056 ms
64 bytes from 10.0.0.1: icmp_seq=5 ttl=64 time=0.050 ms
64 bytes from 10.0.0.1: icmp_seq=6 ttl=64 time=0.082 ms

--- 10.0.0.1 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 508ms
rtt min/avg/max/mdev = 0.045/0.284/1.418/0.507 ms
root@mininet-vm:/home/mininet#
```

```
ping: usage error: Destination address required
root@mininet-vm:/home/mininet# ping -c 6 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=5.05 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.158 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.078 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.075 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.052 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.074 ms

--- 10.0.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5081ms
rtt min/avg/max/mdev = 0.052/0.914/5.047/1.848 ms
root@mininet-vm:/home/mininet#
```

Figure 2: Пингование

На h1 добавил 10% потерь пакетов

```
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem loss 10%
root@mininet-vm:/home/mininet# ping -c 100 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.217 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.084 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.163 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.130 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.079 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.049 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.095 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.088 ms
...
--- 10.0.0.2 ping statistics ---
100 packets transmitted, 90 received, 10% packet loss, time 101383ms
rtt min/avg/max/mdev = 0.039/0.071/0.217/0.026 ms
root@mininet-vm:/home/mininet#
```

Figure 3: На h1 10% потерь пакетов

На h2 тоже добавил 10% потерь пакетов

```
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h2-eth0 root netem loss 10%
root@mininet-vm:/home/mininet#

64 bytes from 10.0.0.2: icmp_seq=98 ttl=64 time=0.051 ms
64 bytes from 10.0.0.2: icmp_seq=99 ttl=64 time=0.076 ms
64 bytes from 10.0.0.2: icmp_seq=100 ttl=64 time=0.109 ms

--- 10.0.0.2 ping statistics ---
100 packets transmitted, 78 received, 22% packet loss, time 101359ms
rtt min/avg/max/mdev = 0.045/0.104/1.227/0.145 ms
root@mininet-vm:/home/mininet#
```

Figure 4: На h2 тоже 10% потерь пакетов

Добавил на интерфейсе узла h1 коэффициент потери пакетов 50%

```
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem loss 50  
% 50%
```

```
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=2.75 ms  
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.618 ms  
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.138 ms  
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.058 ms  
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.082 ms  
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.074 ms  
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.078 ms  
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.068 ms  
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.074 ms  
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.052 ms  
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=0.050 ms  
64 bytes from 10.0.0.2: icmp_seq=23 ttl=64 time=0.063 ms  
64 bytes from 10.0.0.2: icmp_seq=25 ttl=64 time=0.232 ms  
64 bytes from 10.0.0.2: icmp_seq=28 ttl=64 time=0.051 ms  
64 bytes from 10.0.0.2: icmp_seq=29 ttl=64 time=0.047 ms  
64 bytes from 10.0.0.2: icmp_seq=30 ttl=64 time=0.067 ms  
64 bytes from 10.0.0.2: icmp_seq=31 ttl=64 time=0.059 ms  
64 bytes from 10.0.0.2: icmp_seq=32 ttl=64 time=0.066 ms  
64 bytes from 10.0.0.2: icmp_seq=33 ttl=64 time=0.054 ms  
64 bytes from 10.0.0.2: icmp_seq=40 ttl=64 time=0.055 ms  
64 bytes from 10.0.0.2: icmp_seq=41 ttl=64 time=0.096 ms  
64 bytes from 10.0.0.2: icmp_seq=42 ttl=64 time=0.077 ms  
64 bytes from 10.0.0.2: icmp_seq=43 ttl=64 time=0.052 ms  
  
--- 10.0.0.2 ping statistics ---  
50 packets transmitted, 23 received, 54% packet loss, time 50154ms  
rtt min/avg/max/mdev = 0.047/0.215/2.752/0.553 ms  
root@mininet-vm:/home/mininet#
```

Figure 5: На интерфейсе узла h1 коэффициент потери пакетов 50%

Добавил на интерфейсе узла h1 0,01% повреждения пакетов

```
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem corrupt 0.01%

root@mininet-vm:/home/mininet# iperf3 -s
warning: this system does not seem to support IPv6 - trying IPv4
-----
Server listening on 5201
-----

root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 58214 connected to 10.0.0.2 port 5201
[ ID] Interval           Transfer     Bitrate      Retr  Cwnd
[ 7] 0.00-1.00 sec      2.27 GBytes  19.5 Gbits/sec    5   690 KBytes
[ 7] 1.00-2.00 sec      2.10 GBytes  18.1 Gbits/sec    6   694 KBytes
[ 7] 2.00-3.00 sec      2.42 GBytes  20.8 Gbits/sec    5   908 KBytes
[ 7] 3.00-4.00 sec      2.35 GBytes  20.2 Gbits/sec    5   947 KBytes
[ 7] 4.00-5.00 sec      2.14 GBytes  18.3 Gbits/sec    4   793 KBytes
[ 7] 5.00-6.00 sec      2.26 GBytes  19.4 Gbits/sec    4   576 KBytes
```

Figure 6: На интерфейсе узла h1 0,01% повреждения пакетов

25% пакетов - сразу, остальные - с задержкой

```
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem delay 1 0ms reorder 25% 50% ☆
root@mininet-vm:/home/mininet# ping -c 20 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=1.45 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=11.3 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=11.0 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=12.2 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=10.4 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=11.9 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=10.4 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=10.9 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=10.6 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=11.0 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=10.4 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=10.8 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=11.1 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=10.8 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=10.9 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=10.6 ms
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=11.0 ms
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=10.8 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=10.6 ms
64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=10.2 ms

--- 10.0.0.2 ping statistics ---
20 packets transmitted, 20 received, 0% packet loss, time 19047ms
rtt min/avg/max/mdev = 1.445/10.422/12.163/2.112 ms
```

Figure 7: 25% пакетов - сразу, остальные - с задержкой

h1 задал правило с дублированием 50% пакетов

```
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem duplica
te 50%
root@mininet-vm:/home/mininet# ping -c 20 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.172 ms
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.178 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.049 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.057 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.058 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.062 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.063 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.085 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.064 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.052 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.052 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.047 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.049 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.067 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.088 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.088 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.053 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=0.052 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=0.083 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=0.081 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=0.082 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=0.079 ms
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=0.049 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=0.062 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=0.063 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=0.128 ms
```

Figure 8: Правило с дублированием 50% пакетов

Написал скрипт

```
mininet@mininet-vm: ~/work/lab_netem_ii/simple-drop
GNU nano 4.8 lab_netem_ii.py

info('*** Adding controller\n')
net.addController('c0')

info('*** Adding hosts\n')
h1 = net.addHost('h1', ip='10.0.0.1')
h2 = net.addHost('h2', ip='10.0.0.2')

info('*** Adding switch\n')
s1 = net.addSwitch('s1')

info('*** Creating links\n')
net.addLink(h1, s1)
net.addLink(h2, s1)

info('*** Starting network\n')
net.start()

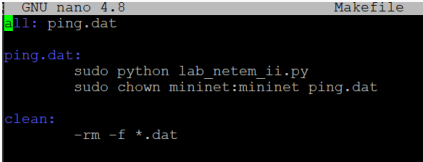
info('*** Setting packet loss 10% on both interfaces\n')
h1.cmdPrint('tc qdisc add dev h1-eth0 root netem loss 10%')
h2.cmdPrint('tc qdisc add dev h2-eth0 root netem loss 10%')

time.sleep(2) # Wait 2 seconds

info('*** Running ping test\n')
# ^r^кполн^оем ping и ^ao^e^`ан^оем полн^кй в^к
ping_output = h1.cmd('ping -c 20 ' + h2.IP())
# ^r^квводим полн^с^н ин^do^`ма^фи^н о ping на ^m
info('*** Ping results:\n')
print("=" * 50)
print("PING TEST RESULTS:")
print("=" * 50)

^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos
```

Создал Makefile

A screenshot of a terminal window showing the GNU nano 4.8 text editor editing a file named 'Makefile'. The editor's interface includes a title bar with 'GNU nano 4.8' and 'Makefile'. The content of the file is as follows:

```
all: ping.dat

ping.dat:
    sudo python lab_netem_ii.py
    sudo chown mininet:mininet ping.dat

clean:
    -rm -f *.dat
```

Figure 10: Makefile

Запуск make

```
0.=====
PING TEST RESULTS:
=====
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.604 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.218 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.069 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.109 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.066 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.113 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.061 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.051 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.075 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=1.35 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.068 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=0.047 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=0.080 ms
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=0.071 ms
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=0.045 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=0.089 ms
64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=0.063 ms

--- 10.0.0.2 ping statistics ---
20 packets transmitted, 17 received, 15% packet loss, time 19419ms
rtt min/avg/max/mdev = 0.045/0.187/1.353/0.318 ms

=====
*** Saving filtered data to ping.dat
```

Figure 11: Запуск make

Написал lab_netem_loss.py

```
GNU nano 4.8 lab_netem_loss.py

# Create hosts
h1 = net.addHost('h1', ip='10.0.0.1')
h2 = net.addHost('h2', ip='10.0.0.2')
s1 = net.addSwitch('s1')

# Create links
net.addLink(h1, s1)
net.addLink(h2, s1)

# Start network
net.start()

info('*** Setting 15% packet loss on h1\n')
h1.cmd('tc qdisc add dev h1-eth0 root netem loss 15%')

time.sleep(2)

info('*** Running ping test with packet loss\n')

# Run ping and display full results
ping_output = h1.cmd('ping -c 30 10.0.0.2')

print("=" * 60)
print("PACKET LOSS EXPERIMENT RESULTS:")
print("=" * 60)
print(ping_output)
print("=" * 60)

# Save detailed timing data
h1.cmd('ping -c 30 10.0.0.2 | grep "time=" > ping_loss.log')

# Display packet loss statistics
loss_stats = h1.cmd('ping -c 30 10.0.0.2 | grep "packet loss"')
```

^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos
^Y Exit ^R Read File ^\ Replace ^U Paste Text ^T To Spell ^_ Go To Lin

Запуск lab_netem_loss.py

```
*** Setting 15% packet loss on h1
*** Running ping test with packet loss
=====
PACKET LOSS EXPERIMENT RESULTS:
=====
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=1.75 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.277 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.049 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.060 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.059 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.051 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.092 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=0.056 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=0.081 ms
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=0.105 ms
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=0.060 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=0.066 ms
64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=0.050 ms
64 bytes from 10.0.0.2: icmp_seq=21 ttl=64 time=0.086 ms
64 bytes from 10.0.0.2: icmp_seq=22 ttl=64 time=0.059 ms
64 bytes from 10.0.0.2: icmp_seq=23 ttl=64 time=0.059 ms
64 bytes from 10.0.0.2: icmp_seq=24 ttl=64 time=0.074 ms
64 bytes from 10.0.0.2: icmp_seq=25 ttl=64 time=0.061 ms
64 bytes from 10.0.0.2: icmp_seq=26 ttl=64 time=0.081 ms
64 bytes from 10.0.0.2: icmp_seq=28 ttl=64 time=0.078 ms
64 bytes from 10.0.0.2: icmp_seq=29 ttl=64 time=0.052 ms
64 bytes from 10.0.0.2: icmp_seq=30 ttl=64 time=0.057 ms

--- 10.0.0.2 ping statistics ---
30 packets transmitted, 25 received, 16.6667% packet loss, time 29639ms
rtt min/avg/max/mdev = 0.049/0.141/1.747/0.330 ms
```

Написал lab_netem_reorder.py

```
GNU nano 4.8 lab_netem_reorder.py
#!/usr/bin/env python

"""
Experiment with packet reordering
"""

from mininet.net import Mininet
from mininet.node import Controller
from mininet.log import setLogLevel, info
import time

def reorder_experiment():
    "Experiment with packet reordering"

    net = Mininet(controller=Controller, waitConnected=True)
    net.addController('c0')

    # Create hosts
    h1 = net.addHost('h1', ip='10.0.0.1')
    h2 = net.addHost('h2', ip='10.0.0.2')
    s1 = net.addSwitch('s1')

    # Create links
    net.addLink(h1, s1)
    net.addLink(h2, s1)

    # Start network
    net.start()

    info('*** Setting packet reordering: 25% packets reordered with 50ms delay\n')
    h1.cmd('tc qdisc add dev h1-eth0 root netem delay 50ms reorder 25%')

    time.sleep(2)

[ Read 67 lines ]
^G Get Help      ^O Write Out    ^W Where Is     ^K Cut Text     ^J Justify      ^C Cur Pos
^X Exit          ^R Read File    ^\ Replace      ^U Paste Text   ^T To Spell     ^_ Go To L
```

Запуск lab_netem_reorder.py

```
$1
*** Setting packet reordering: 25% packets reordered with 50ms delay
*** Running ping test with packet reordering
=====
PACKET REORDERING EXPERIMENT RESULTS:
=====
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=51.8 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=50.6 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=50.6 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=50.3 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.077 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.065 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=50.3 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=50.5 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=50.3 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.097 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=50.3 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=50.4 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=50.9 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=50.7 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=50.2 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=50.5 ms
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=0.075 ms
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=51.0 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=51.6 ms
64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=50.5 ms

--- 10.0.0.2 ping statistics ---
20 packets transmitted, 20 received, 0% packet loss, time 19127ms
rtt min/avg/max/mdev = 0.065/40.545/51.817/20.237 ms
```

Figure 15: 3anyck lab_netem_reorder.py

Написал lab_netem_corruption.py

```
#!/usr/bin/env python

"""
Experiment with packet corruption
"""

from mininet.net import Mininet
from mininet.node import Controller
from mininet.log import setLogLevel, info
import time

def corrupt_experiment():
    "Experiment with packet corruption"

    net = Mininet(controller=Controller, waitConnected=True)
    net.addController('c0')

    # Create hosts
    h1 = net.addHost('h1', ip='10.0.0.1')
    h2 = net.addHost('h2', ip='10.0.0.2')
    s1 = net.addSwitch('s1')

    # Create links
    net.addLink(h1, s1)
    net.addLink(h2, s1)

    # Start network
    net.start()

    info('*** Setting 1% packet corruption on h1\n')
    h1.cmd('tc qdisc add dev h1-eth0 root netem corrupt 1%')

    time.sleep(2)
```

^G Get Help

^X Exit

^O Write Out

^R Read File

^W Where Is

^_ Replace

[Read 73 lines]

^K Cut Text

^U Paste Text

^J Jus

^T To

Запуск lab_netem_corruption.py

```
IPERF3 TEST (10 seconds):
Connecting to host 10.0.0.2, port 5201
[ 5] local 10.0.0.1 port 58236 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer    Bitrate      Retr  Cwnd
[ 5]  0.00-1.00    sec  1.35 GBytes  11.6 Gbits/sec  548   84.8 KBytes
[ 5]  1.00-2.00    sec  1.06 GBytes  9.13 Gbits/sec  406   161 KBytes
[ 5]  2.00-3.00    sec  1.42 GBytes  12.2 Gbits/sec  465   144 KBytes
[ 5]  3.00-4.00    sec  1.32 GBytes  11.3 Gbits/sec  439   168 KBytes
[ 5]  4.00-5.00    sec  1.18 GBytes  10.2 Gbits/sec  392   105 KBytes
[ 5]  5.00-6.00    sec  1.33 GBytes  11.4 Gbits/sec  432   59.4 KBytes
[ 5]  6.00-7.00    sec  1.13 GBytes  9.68 Gbits/sec  369   247 KBytes
[ 5]  7.00-8.00    sec  1.07 GBytes  9.23 Gbits/sec  370   720 KBytes
[ 5]  8.00-9.00    sec  1.40 GBytes  12.0 Gbits/sec  450   321 KBytes
[ 5]  9.00-10.00   sec  1.11 GBytes  9.51 Gbits/sec  362   202 KBytes
-----
[ ID] Interval      Transfer    Bitrate      Retr
[ 5]  0.00-10.00   sec  12.4 GBytes  10.6 Gbits/sec  4233
[ 5]  0.00-10.00   sec  12.4 GBytes  10.6 Gbits/sec
                                     sender
                                     receiver

iperf Done.
```

Figure 17: Запуск lab_netem_corruption.py

Получил навыки проведения интерактивных экспериментов в среде Mininet по исследованию параметров сети, связанных с потерей, дублированием, изменением порядка и повреждением пакетов при передаче данных.